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Development of the RFM Model for Predicting Customer Lifetime Value Using a Hybrid Clustering and Stacked Ensemble Learning Model in the Fast-Moving Consumer Goods (FMCG) Industry

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A Thesis Presented in Partial Fulfillment of the Requirements for the Degree of Master of Science in

Industrial Engineering

Mar 2025

Abstract:

In today's competitive world, predicting Customer Lifetime Value (CLV) is recognized as a key tool for optimizing marketing decisions and resource allocation, and identifying this value is highly important for business owners. Various methods have been proposed so far for predicting and estimating this value; this research aims to propose an innovative and hybrid model and validate it for CLV prediction. This study, with the goal of developing a hybrid machine learning model based on clustering and regression, provides an innovative framework for predicting customer lifetime value. In this model, customer features are extracted using an advanced RFM approach, and then the k-means algorithm is used for clustering customer behavior. The clustering results are used as inputs to powerful machine learning models, including Gradient Boosting Regression, Random Forest Regression, and Poisson Regression, as base learners. By utilizing the ensemble learning algorithm, stacked regression and ridge regression are employed as meta-models, improving the prediction accuracy compared to the base models.

The proposed model demonstrated significant performance in prediction with an R^2 value of 0.91 and a Mean Absolute Error (MAE) of 57.13. This study, focusing on the Fast-Moving Consumer Goods (FMCG) industry and real customer data, provides a practical and implementable framework for strategic decision-making. The results of this research can help businesses optimize their marketing strategies, reduce advertising costs, and increase return on investment. Furthermore, the proposed model will serve as a foundation for developing more advanced customer lifetime value prediction systems in various industries.

Keywords:

Customer Behavior Analysis, Customer Lifetime Value, Machine Learning, Ensemble Learning